

THOMASVILLE 2 S, AL, USA

WMO: 721551

Lat: 31.8814N

Lon: 87.7367W

Elev: 350

StdP: 14.51

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 23802

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.9	(c) 26.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.7	(k) 62.5	(l) 13.0	(m) 56.9	(n) 3.8	(o) 350	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.1	(c) 94.3	(d) N/A	(e) 92.3	(f) N/A	(g) 90.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.3	(p) 350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.3	(b) 11.6	(c) 10.1	(e) 17.1	(f) 96.3	(g) 4.9	(h) 3.1	(i) 13.5	(j) 98.5	(k) 10.6	(l) 100.3	(m) 7.9	(n) 102.1	(o) 4.3	(p) 104.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.4	46.0	51.6	57.3	64.1	71.3	78.4	79.9	79.3	75.7	64.7	54.2	50.1		
	DBStd	13.99	10.78	10.14	9.00	6.57	5.63	3.54	2.72	3.19	4.78	7.68	8.42	9.87		
	HDD50	524	207	98	37	1	0	0	0	0	0	3	50	128		
	HDD65	2269	591	388	264	95	19	0	0	0	1	101	339	471		
	CDD50	5795	84	141	264	425	660	852	925	908	771	460	174	131		
	CDD65	2065	3	12	26	69	214	402	460	443	322	92	13	9		
	CDH74	18093	9	46	183	630	1847	3808	4162	3809	2676	838	69	16		
	CDH80	7366	0	1	17	95	611	1744	1886	1696	1112	198	6	0		
(14) Wind	WSAvg	4.1	4.8	5.7	5.8	4.6	3.6	3.5	3.3	3.3	2.9	3.5	3.5	4.8		
(15) Precipitation	PrecAvg	59.1	5.7	5.2	6.7	4.3	4.3	4.7	6.1	4.4	4.1	3.4	4.6	5.4		
	PrecMax	83.3	12.2	14.3	15.4	12.0	12.0	12.7	15.9	9.9	13.4	10.4	11.3	18.2		
	PrecMin	40.0	1.1	1.3	1.3	0.5	0.0	0.2	0.7	1.0	0.1	0.0	0.9	1.1		
	PrecStd	10.0	2.5	2.7	3.6	2.6	2.6	2.9	3.1	2.5	3.0	2.8	2.7	3.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	78.8	82.3	84.6	91.3	96.2	95.6	95.8	94.8	87.4	80.2	76.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	71.3	75.0	79.1	82.7	88.4	93.9	93.1	93.5	91.7	84.9	75.6	73.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	67.4	71.3	75.5	80.8	86.0	91.5	91.3	91.0	89.3	82.2	72.7	70.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	63.6	68.0	72.1	78.1	83.8	89.1	89.4	88.8	86.8	79.2	69.5	67.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.4	21.8	22.8	23.5	21.9	20.5	19.1	18.9	20.4	24.0	24.1	19.5		
		MCDBR	22.2	22.1	23.7	25.0	24.9	23.6	21.7	22.1	23.7	25.9	25.3	19.6		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.337	0.361	0.388	0.434	0.461	0.523	0.514	0.439	0.374	0.344	0.326		
	taud		2.544	2.502	2.438	2.355	2.264	2.236	2.077	2.098	2.289	2.447	2.482	2.539		
	Ebn at Noon		287	292	292	286	272	264	247	248	263	275	276	279		
	Edh at Noon		27	30	35	39	43	45	52	50	40	32	28	26		
(44) All-Sky Solar Radiation	RadAvg		890	1069	1438	1786	1963	1957	1884	1762	1590	1353	1017	775		
	RadStd		68	143	128	145	127	157	118	108	115	161	96	78		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+266	+173

Nomenclature: See separate page